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PI Name:	Paik, Ho Jung Ph.D.		
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Key Personnel Changes/Previous PI:	Paik, Ho Jung		
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	The test Newton's law of gravitation with a resolution one million times better than the existing limits at 0.1 mm, the superconducting accelerometer technology will be combined with advantages of the low-g environment of space. The experiment will be cooled to < 2 K and the test masses will be magnetically suspended. The soft, low-loss suspension, combined with a low-noise SQUID, leads to extremely low intrinsic noise in the detector. To minimize Newtonian errors, the experiment employs a near null source, a circular disk of large diameter-to-thickness ratio. Two test masses, also disk-shaped, are suspended on the two sides of the source mass at a distance of 0.1 mm and form a superconducting differential accelerometer. A ground test apparatus is under construction and a laboratory test of the inverse-square law is planned. The objective of ISLES (Inverse-Square Law Experiment in Space) is to perform a null test of Newton's law on ISS with		

Task Description:

a resolution of 10 ppm at distances ranging from 100 micrometer to 1 mm. This represents an improvement by 10^6 over the existing limits at 100 micrometers, and even greater improvement at shorter distances. ISLES will be sensitive enough to detect axions with the strongest allowed coupling and test the string-theory prediction with R > 5 micrometer. To reach this sensitivity, the existing superconducting accelerometer technology will be combined with advantages of the low-g environment of space.

The Standard Model of particle physics, although highly successful, has one serious blemish: the so-called "strong CP problem." A very plausible explanation for this phenomenon was proposed over two decades ago, but experimentalists have so far failed to verify it. One ramification of the theory is the prediction of the existence of a new particle, called the "axion," which mediates a short-range mass-mass interaction, as well as a force between intrinsic spin and mass. The axion is also a strong candidate for cold dark matter. Confirmation (or rejection) of this prediction would therefore have a major impact on our understanding of the universe, from its most microscopic constituents to its grand structure.

Another impetus for testing the law at the shortest ranges possible comes from string theories. In these theories, the extra spacetime dimensions are compactified, leaving the observed 3+1 spacetime dimensions. For two large extra dimensions, the law of gravity changes from 1/r^2 to 1/r^4, as the separation r is reduced to below the radius of compactification R. An intriguing possibility that R > 1 micrometer has been suggested recently that would imply a violation of the inverse-square law below 1 micrometer distance. This is the first prediction of a string theory that can be tested, and a discovery of this deviation would be ground breaking.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The superconducting accelerometer technology developed in this project will have applications in exploration of natural resources, detection of hidden masses, and inertial navigation.

Task Progress:

A. Design of the apparatus
To eliminate differential contraction and provide good electromagnetic shielding, the entire housing is fabricated from niobium (Nb). The source mass is made out of tantalum (Ta), which closely matches Nb in thermal contraction. It is suspended by cantilever springs and driven magnetically. The test masses, also made out of Ta, are suspended by cantilever springs and coupled magnetically to form a differential accelerometer. A thin Nb sheet provides electrostatic and magnetic shielding between the source and each test mass.

Since the source mass shares the same housing as the test masses, special provisions must be made to prevent the reaction forces from distorting the detector. To this end, the inner parts of the housing containing the detector are decoupled from the outer rim by inserting a mechanical weak link, a cantilever spring, between them. This soft suspension also permits the alignment of the test masses. (The soft suspension of the source mass, however, amplified disturbances from the source mass, as will be discussed in Section E.)

As the source mass is driven at frequency f along the symmetry axis, the first-order Newtonian fields arising from the finite diameter of the source mass are canceled upon differential meas-urement, leaving only a second-order error at 2f. By symmetry, a violation signal also appears at 2f. The second harmonic detection, combined with the CMRR (Common Mode Rejection Ratio) of the detector, reduces vibration coupling by over 200 dB.

B. Analysis of metrology errors

A detailed analysis has been carried out for source metrology errors to determine the required machining tolerances. The azimuthal density and dimensional errors can be averaged out by rotating the test masses. The radial density variation was expanded as a Fourier Bessel series with random coefficients. The thickness variation was represented as random surface density fluctuations. In spite the proximity of the test mass to the source mass (~ 0.1 mm), high-order terms of up to order 100, which corresponds to a resolution of ~ 4 mm, contribute significantly to the total error. The thickness fluctuations are much more important than the density error.

C. Analysis and design of the superconducting circuits

The driving and sensing circuits were fully analyzed to optimize their performance and define the inductances needed for the pancake coils and superconducting transformers. Detailed examination and modeling of the source driving resulted in a change in the circuitry. The new circuit uses much smaller inductors and permits cancellation of the nonlinear term, which is important to avoid generating an undesirable Newtonian force at the second harmonic. To reduce the recoil of the apparatus caused by the source motion and the resulting common-mode acceleration, a force will be applied from the cryostat to the suspended housing. For this purpose, two shaker coils have been added on the Nb covers, as well as a shaker driving circuit.

D. Construction of the apparatus

Two 1.6-mm thick Ta sheets were "two face grounded" commercially to improve the surface flatness. A measurement with dial indicators showed the surface to be flat to 10 microns, which is an acceptable tolerance for the ground experiment. A source mass was produced from one of these sheets by wire EDM and then heat-treated to relieve stress. Four test masses were constructed, also with wire EDM, from a 0.25-mm thick Ta sheet. Two Nb housings that hold the source mass, test masses, and various coils together were machined. A new Nb hot zone had to be constructed for our high-vacuum oven to accommodate the 22.9-cm diameter housings and source mass. Superconducting shields for the test masses were fabricated by diffusion-bonding 12.5-micron Nb sheets to Nb rings. The source driving coils, various sensing coils, and alignment coils were wound. The superconducting transformers were wound and tested. To reduce the susceptibility of the SQUID amplifiers to rf interference, shielded cables for the current charging leads were constructed and added to the cryostat. The apparatus needs to be cooled to 2 K to reduce its sensitivity to temperature fluctuation. The cryostat was cooled down several times with the impedance of the capillary for the 2-K stage varied until a stable operation was achieved near 2 K. The entire apparatus was assembled and integrated with the cryostat. This required careful mechanical alignment and assembly of the source and test masses as well as coils and shields, complicated routing of dozens of leads for various circuits, and making scores of spot-welded superconducting joints.

E. Test of the apparatus

The entire apparatus has been tested extensively, which has led to making a number of corrections to the apparatus. By

adjusting the persistent currents, we were able to obtain a CMRR of 10^4. The differential mode (DM) response at the source driving frequency f was mainly due to magnetic cross-talk, which corresponded to a 120-dB attenuation, much larger than we expected. The DM peak at 2f, at which frequency an inverse-square law violation signal should appear, showed a large Newtonian error, 10^5 times higher than we expected from known error sources. A design flaw in the apparatus magnified disturbances from the source.

In our apparatus, the center parts of the housings, which contain the test masses, were soft suspended from the rim, which supports the source mass. This was done to reduce the DM distortion of the housing at 2f. Unfortunately, as the source mass moves with respect to the test masses, the disturbing forces from patch fields, Casimir force, and residual gas pressure now directly drive the two test mass blocks in DM, short-circuiting the CM rejection. To make these effects CM and reject them by 10^4, the apparatus must be modified. The source mass could be suspended and driven from outside the housings. The pendulum mode for the source mass will be ~1 Hz. The Nb flexures act as a negative spring and reduce the resonance frequency further. This low-frequency suspension (compared to present 13 Hz) gives a number of advantages. The compliant source can be driven with a much smaller magnetic field. By driving the source mass at its resonance, the driving field can be reduced further by a factor of its Q (~10^5). This will virtually eliminate the magnetic cross-talk problem. The resonance experiment enables driving the source at a single frequency, which is otherwise very difficult due to the nonlinearity of the source driving circuit. This eliminates the nonlinearity-driven second-harmonic, another very important error source. We are currently building a prototype to test this new source suspension.

This ground research has led to a much better understanding of the error sources for the inverse-square law experiment, and has given us much insight into designing an improved, ground and space experiments.

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